Supplementary Table 2. Top 100 signature genes of malignant cells versus non-malignant cells at single cell level

Gene	Log(Fold-change)	Pct.1	Pct.2	*p*-value	Adj *p*-value
REG4	2.612727632	0.64	0.1	0.00E+00	0.00E+00
TFF3	2.516402776	0.59	0.11	0.00E+00	0.00E+00
PHGR1	2.490764167	0.75	0.219	0.00E+00	0.00E+00
FABP1	2.425470859	0.42	0.014	0	0
PRAP1	2.351212197	0.57	0.025	0	0
APOA1	2.336508716	0.19	0.058	4.0328E-100	7.41559E-96
CLDN4	2.034301452	0.82	0.097	0.00E+00	0.00E+00
LEFTY1	1.946601555	0.34	0.007	0.00E+00	0.00E+00
SPINK4	1.86874387	0.32	0.013	0.00E+00	0.00E+00
ANPEP	1.841039463	0.36	0.012	0	0
CLDN7	1.606715566	0.76	0.041	0.00E+00	0.00E+00
ALDOB	1.554569816	0.3	0.01	0.00E+00	0.00E+00
S100A1(	1.419158923	0.87	0.5	0.00E+00	0.00E+00
IL32	1.371711372	0.63	0.094	0.00E+00	0.00E+00
MALAT1	1.328960584	1	0.989	0	0
LGALS4	1.316374911	0.82	0.507	0.00E+00	0.00E+00
CLDN3	1.282105814	0.51	0.022	0.00E+00	0.00E+00
MDK	1.241192709	0.59	0.208	0	0
OLFM4	1.13099641	0.29	0.085	4.2693E-162	7.8504E-158
PRSS3	1.112233633	0.61	0.068	0	0
LGALS3	1.110261325	0.8	0.742	3.9794E-179	7.3173E-175
DMBT1	1.094236054	0.28	0.015	3.14E-304	5.78E-300
CES2	1.026709291	0.58	0.153	0.00E+00	0.00E+00
HLA-A	0.998443111	0.93	0.849	0.00E+00	0.00E+00
CDHR5	0.990555325	0.3	0.055	6.12E-224	1.13E-219
FABP2	0.948062893	0.2	0.003	4.25E-226	7.81E-222
CCT2	0.942218569	0.45	0.266	7.80E-144	1.43E-139
PPP1R1	0.918932882	0.68	0.341	6.9894E-297	1.2852E-292
AOC1	0.917461431	0.45	0.047	0.00E+00	0.00E+00
TM4SF2	0.896324787	0.32	0.005	0.00E+00	0.00E+00
KRT7	0.871428496	0.44	0.096	0.00E+00	0.00E+00
NFKBIA	0.866391666	0.64	0.45	1.05E-136	1.93E-132
ISG15	0.841077295	0.48	0.177	4.77E-229	8.78E-225
CCL20	0.834419708	0.2	0.021	2.49E-179	4.57E-175
C8orf4	0.794764954	0.43	0.153	1.16E-212	2.13E-208
FTL	0.786086622	0.92	0.916	7.03E-69	1.29E-64
AGR3	0.785856065	0.61	0.311	3.22E-271	5.91E-267
CXCL1	0.777644237	0.29	0.045	2.20E-240	4.04E-236
RBP2	0.776760648	0.11	0.005	2.97E-110	5.46E-106
TMC5	0.769922572	0.7	0.433	2.03E-247	3.73E-243
PCK1	0.757825693	0.26	0.004	6.72E-303	1.23E-298
TRIM31	0.749418465	0.44	0.079	0.00E+00	0.00E+00
CDH17	0.747362417	0.39	0.011	0.00E+00	0.00E+00
PI3	0.745585921	0.23	0.03	1.28E-190	2.35E-186
HES1	0.73760363	0.69	0.534	1.26E-104	2.31E-100

LCN2	0.733063095	0.52	0.273	1.02E-149	1.88E-145
TMEM17	0.708261903	0.39	0.03	0.00E+00	0.00E+00
TM4SF4	0.707157903	0.22	0.017	1.65E-214	3.03E-210
SERINC2	0.703530511	0.57	0.282	7.23E-275	1.33E-270
MISP	0.702219402	0.52	0.155	0.00E+00	0.00E+00
REP15	-0.67148319	0.04	0.325	0.00E+00	0.00E+00
TM9SF3	-0.679718736	0.48	0.726	0.00E+00	0.00E+00
SLC12A2	-0.679903188	0.23	0.565	0.00E+00	0.00E+00
IL33	-0.682804676	0.03	0.371	0.00E+00	0.00E+00
RPS4Y1	-0.70616236	0.36	0.562	3.25E-193	5.97E-189
BRI3	-0.712467886	0.38	0.652	1.42E-299	2.61E-295
GHRL	-0.726686611	0.07	0.412	0.00E+00	0.00E+00
S100P	-0.728083876	0.82	0.852	1.36E-208	2.51E-204
CXCL17	-0.739067791	0.27	0.764	0.00E+00	0.00E+00
PAPSS1	-0.742213208	0.08	0.454	0.00E+00	0.00E+00
FOXQ1	-0.75564486	0.22	0.541	0.00E+00	0.00E+00
FABP5	-0.765226992	0.43	0.645	2.32E-233	4.27E-229
CD63	-0.771503923	0.68	0.901	0.00E+00	0.00E+00
CLDN18	-0.777709594	0.47	0.891	0.00E+00	0.00E+00
MUC1	-0.778888762	0.68	0.952	0.00E+00	0.00E+00
KLF2	-0.80051594	0.22	0.476	5.08E-221	9.35E-217
CTSE	-0.803818982	0.44	0.83	0.00E+00	0.00E+00
SOSTDC	-0.807940176	0.01	0.357	0.00E+00	0.00E+00
EEF1A1	-0.819681969	0.94	0.984	0.00E+00	0.00E+00
ANXA10	-0.83075542	0.45	0.854	0.00E+00	0.00E+00
KCNE2	-0.849158333	0.01	0.432	0.00E+00	0.00E+00
TAGLN2	-0.862507052	0.62	0.861	0.00E+00	0.00E+00
TSPO	-0.921784021	0.64	0.871	0.00E+00	0.00E+00
VSIG1	-0.926687654	0.24	0.68	0.00E+00	0.00E+00
AGR2	-0.936868786	0.87	0.982	0.00E+00	0.00E+00
METTL7	-0.96905016	0.12	0.611	0.00E+00	0.00E+00
ALDH1A	-1.00068719	0.35	0.738	0.00E+00	0.00E+00
VSIG2	-1.074448236	0.44	0.861	0.00E+00	0.00E+00
PSCA	-1.082273135	0.47	0.894	0.00E+00	0.00E+00
ALDH3A	-1.119698127	0.13	0.596	0.00E+00	0.00E+00
PGA4	-1.134841003	0	0.209	1.10E-280	2.03E-276
TMSB4X	-1.170442574	0.97	1	0.00E+00	0.00E+00
DPCR1	-1.245947673	0.21	0.661	0.00E+00	0.00E+00
SPINK1	-1.268455701	0.62	0.818	0.00E+00	0.00E+00
IGFBP2	-1.315513043	0.13	0.745	0.00E+00	0.00E+00
CA9	-1.33110057	0.15	0.711	0.00E+00	0.00E+00
GLUL	-1.351117423	0.21	0.746	0.00E+00	0.00E+00
PGA3	-1.362343716	0.03	0.326	0.00E+00	0.00E+00
TFF1	-1.36505636	0.71	0.867	2.97E-265	5.46E-261
MSMB	-1.456155777	0.25	0.573	0.00E+00	0.00E+00
CA2	-1.575866278	0.37	0.875	0.00E+00	0.00E+00
CYSTM1	-1.639388365	0.78	0.985	0.00E+00	0.00E+00
RNASE1	-1.706656562	0.22	0.887	0.00E+00	0.00E+00

MUC6	-1.788053329	0.1	0.293	3.03E-164	5.58E-160
MUC5AC	-2.014531322	0.56	0.908	0.00E+00	0.00E+00
TFF2	-2.434654634	0.52	0.919	0.00E+00	0.00E+00
GKN2	-2.642135586	0.09	0.615	0.00E+00	0.00E+00
GKN1	-3.128042381	0.16	0.527	0.00E+00	0.00E+00
PGC	-4.159308631	0.25	0.88	0.00E+00	0.00E+00
LIPF	-4.744955989	0.2	0.802	0.00E+00	0.00E+00